Functions & Graphs

Section 1: Rational Functions & Oblique Asymptotes

1 Find the oblique asymptote of $f(x) = \frac{3x^2 - 2x + 4}{x - 1}$.

a. $y = 3x + 1$

b. $y = 3x - 1$

c. $y = 3x + 4$

d. $y = 3x$

2 Where does $g(x) = \frac{x^2 - 16}{x^2 - 3x - 4}$ have a vertical asymptote?

a. $x = -1$

b. $x = 4$

c. $x = -4$

d. Both $x = -1$ and $x = 4$

3 What is the horizontal asymptote of $h(x) = \frac{5x - 2}{x^2 + 1}$?

a. $y = 0$

b. $y = 5$

c. $y = -2$

d. No horizontal asymptote

Section 2: Piecewise & Absolute Value Graphs

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4 What is the range of $f(x) = |2x - 6| - 4$?

a. $[-4, \infty)$

b. $[4, \infty)$

c. $[0, \infty)$

d. $(-\infty, -4]$

5 At what value of x does the graph of $y = |x - 3| + |x + 5|$ attain its minimum value?

a. Any $x \in [-5, 3]$

b. $x = -1$

c. $x = 3$

d. $x = -5$

6 What is the domain of $f(x) = \sqrt{9 - x^2}$?

a. $[-3, 3]$

b. $(-\infty, -3] \cup [3, \infty)$

c. $[0, 3]$

d. $[-9, 9]$

Section 3: Inverse Trigonometric Graphs & Domains

7 What is the domain of $f(x) = \arcsin(2x - 1)$?

a. $[0, 1]$

b. $[-1, 1]$

c. $[-0.5, 0.5]$

d. $[0, 2]$

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8 What is the range of the principal $\arccos(x)$ function?

a. $[0, \pi]$

b. $[-\frac{\pi}{2}, \frac{\pi}{2}]$

c. $[0, 2\pi]$

d. $(0, \pi)$

9 Evaluate $\cos(\arcsin(-\frac{4}{5}))$.

a. $\frac{3}{5}$

b. $-\frac{3}{5}$

c. $\frac{4}{5}$

d. $-\frac{4}{5}$

Section 4: Transformations of Non-Linear Functions

10 How is the graph of $y = 2f(-x) + 3$ obtained from $y = f(x)$?

a. Reflect over y-axis, vertical stretch by 2, shift up 3

b. Reflect over x-axis, vertical stretch by 2, shift up 3

c. Reflect over y-axis, horizontal stretch by 2, shift up 3

d. Reflect over origin, shift up 3

11 If the point $(2, 6)$ lies on $y = f(x)$, which point must lie on $y = f(2x) - 1$?

a. $(1, 5)$

b. $(4, 5)$

c. $(1, 7)$

d. $(4, 7)$

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12 Which transformation maps $y = x^3$ to $y = -(x - 1)^3 + 2$?

a. Shift right 1, reflect over x-axis, shift up 2

b. Shift left 1, reflect over x-axis, shift up 2

c. Shift right 1, reflect over y-axis, shift down 2

d. Shift left 1, shift up 2

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